Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[18CE721]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

PRESTRESSED CONCRETE STRUCTURES

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** IS 1343:2012 code books is permitted.

1. a. Define the following terms: **08**

- **i.** Pre-stress
- **ii.** Tendon
- **iii.** Pre-tensioning
- **iv.** Post-tensioning

b. Explain with neat sketch, Freyssinet system of post –tensioning **06**

2. a. A concrete beam having a rectangular section 150x300mm is pre-stressed by a parabolic cable at an eccentricity of 75mm at mid-span towards bottom and eccentricity of 25mm towards top at support sections. The effective pre-stressing force of 350kN. The beam supports a concentrated load of 20kN at center of span in addition to the self-weight with a span of 8m. Find the Short term deflection at center of span under Pre-stress+ Self-weight +Live load. Find also long term deflection if the loss ratio is 0.8 and the creep-coefficient is 1.6 and E_c=38kN/mm^2. **10**

b. List the various types of losses in pre-stressed concrete members. **04**

3. a. A pre-stressed concrete beam of rectangular section of 200x500mm deep is pre-stressed by tendon having an area of 600mm^2 located at 100mm from the soffit of beams. Where f_{ck}=40N/mm^2 ,f_p=1600 N/mm^2 , Estimate the flexural strength for the following case **08**

- **i.** If the beam is pre-tensioned with effective bond
- **ii.** If the beam is post tensioned with effective bond

b. Explain with neat sketches, the different types of flexural failures in PSC beams. **06**

OR

4. a. A post-tensioned beam with unbounded tendons is of rectangular section of width 500mm and effective depth of 1000mm. The cross sectional area of the pre-stressing steel is 3000mm^2. The effective pre-stress after considering all the losses is 1000MPa. The effective span of the beam made of M40 concrete is 15m. Estimate the ultimate moment of resistance of the section using codal provisions. **07**

b. A pre-tensioned T section having flange width of 1200mm and thickness is 150mm, width and depth of rib 300 mm and 1500mm respectively. The area of tensile steel is 4700mm^2 located at an effective depth of 1600mm. If the characteristic compressive strength of concrete and steel are 40N/mm^2 and 1600 N/mm^2.Calculate the flexural strength at limit state of collapse in flexure for the T- Section as per IS code Provisions. **07**

5. a. Explain the types of shear cracks in structural concrete. **04**

b. A concrete beam of rectangular section has a width of 250mm and depth of 600mm. The beam is pre-stressed by a parabolic cable carrying an effective force of 1000kN. The cable is concentric at support and has a maximum eccentricity of 100mm at the center of span. The beam spans over 10m and supports uniformly distributed load of 20kN/m. Assuming the density of concrete as 24kN/m^3, estimate (a) the maximum principle stress developed in the section of the beam at a distance of 300mm from the support, (b) the pre-stressing force required to nullify the shear force to dead and live loads at the support sections. **10**

OR

6. a. Explain different methods of improving the shear resistance in PSC members. **04**

b. The support section of PSC beam 120x250mm is required to carry an ultimate shear force of 70kN. The compressive stress at centriodal axis is 5MPa and f_{ck}=40MPa. f_y= 415 N/mm^2 cover to reinforcement =50mm. Design the shear reinforcement at the section as per IS-1343 Recommendations. **10**

7. a. Explain the concept of anchorage zone stresses. **04**

b. The end block of post-tensioned beam of 500mmx1000mm is pre-stressed with two cables each comprising of five wires of 7mm diameter. The cable is anchored by square plates of 400mmx400mm located at 250mm from the top and bottom edges of the beam. The jacking force in the cable is 3000kN. Design a suitable anchorage zone reinforcement as per IS 1343 recommendations. **10**

OR

8. a. A freyssinet anchorage of 100mm diameter carrying 12 wires of 7mm diameter is embedded concentrically in the web of I-section with web thickness of 225mm. Using IS code method determine the tensile and bursting tensile force in the end blocks. Design the end block and sketch the reinforcement details. **14**